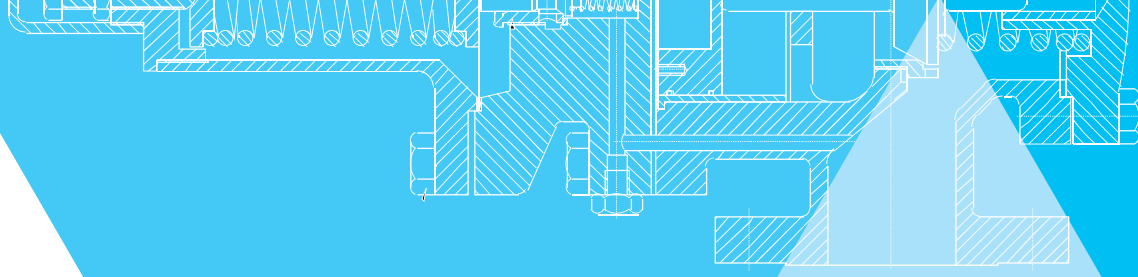
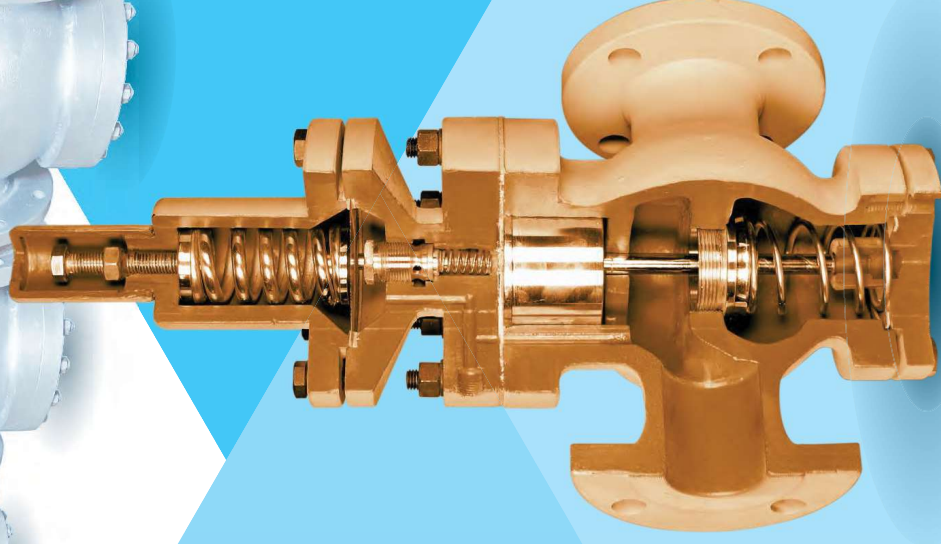


# Pressure Reducing Valve

- ▶ **PRV-100 SERIES**  
(Self Actuating-Mechanical Type)
- ▶ **RCV-100 SERIES**  
(Roboter Operated)
- ▶ **PCV-100 SERIES**  
(Pneumatic / Motorized Operated)

## UNIQUE QUALITIES

- ▶ The "Steam-Con" series PRV-100, Internally Pilot & Piston operated pressure reducing valves are very compact in design & versatile type for accurate regulation with the widest possible range of flow.
- ▶ The full lift type main valve assembly & globe type body design is capable of delivering very high flow rates with greater accuracy.
- ▶ The internally Pilot controlled pressure reducing valves are best suitable for use in continuous distribution application.
- ▶ The constant outlet pressure (downstream pressure) is accurately maintained right up to maximum rated discharge capacity of the valve and is unaffected by changes in inlet pressure or flow rate.
- ▶ A positive tight shut off ensured by accurate guiding of the Pilot & the main valve assembly.
- ▶ RCV-100 Series Roboter operated pressure reducing Valve is high in demand where accurate down stream pressure control is required with high flow.
- ▶ PCV-100 Series Pneumatic / Motorized operated pressure reducing valve is high in demand where accurate down stream high pressure control with high flow is required.
- ▶ Correct selection of material & treatment for longer life of trims and internals.
- ▶ Superior Quality of finished components and best workmanship.
- ▶ 100% spares interchangeability.
- ▶ Different types / models of pressure reducing valves are available as per the different applications required by the industries
- ▶ Each PRV is correctly sized with advance sizing selection software for accurate controlled performance.



# Steam-Con

## Self Actuating-Mechanical Type Pressure Reducing Valve

### PRV-100 SERIES



PRV-100 F

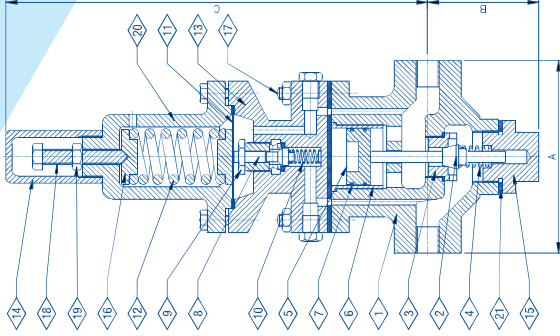
#### Product description:

Self Actuating, Internally Sensed Pilot & Piston operated, Direct Spring & Diaphragm Actuated, Top & Bottom Guided Pressure Reducing Valve with screwed / Flanged End connection.

#### Design:

- PRV-100 Series type PRV consisting of two Valves - an internal Pilot & Main Valve
- Highly Sensitive Pilot
- This Valve Operates on Spring & Diaphragm Balance Principal
- Very High flow Rate due to full lift of main Valve and Body design
- Extremely compact & Light weight design
- The Internal Pilot design eliminates external Piping Sense

PRV-100 T



#### Material Specifications

| Sr.No. | Part Name            | Material                    |
|--------|----------------------|-----------------------------|
| 1      | Body                 | ASTM A216 GR. WCB           |
| 2      | Main Valve           | ASTM A351 GR. C18           |
| 3      | Main Valve Seat      | ASTM A351 GR. C18           |
| 4      | Main Valve Spring    | SS 302                      |
| 5      | Piston               | SS 304 / SS 316             |
| 6      | Sleeve               | ASTM A351 GR.CF8            |
| 7      | Piston Ring          | Hard Chrome Ion             |
| 8      | Pilot Valve          | SS 304 / SS 316             |
| 9      | Pilot Valve Seat     | SS 304 / SS 316             |
| 10     | Pilot Valve Spring   | SS 302                      |
| 11     | Diaphragm            | SS 316                      |
| 12     | Setting Spring       | Carbon Steel                |
| 13     | Pilot Valve Housing  | ASTM A216 GR.WCB            |
| 14     | Cap                  | ASTM A216 GR.WCB            |
| 15     | Bottom Plug/Cover    | M.S./ASTM A216 GR.WCB       |
| 16     | Spring Washer        | Mild Steel                  |
| 17     | Bolts                | ASTM A193 GR.B7             |
| 18     | Adjusting Screw      | M.S./ASTM A193 GR. B7       |
| 19     | Adjusting Screw Nut  | M.S./ASTM A193 GR. 2H       |
| 20     | Spring Housing       | ASTM A216 GR.WCB            |
| 21     | Gasket               | Aramid Fiber (Non-Asbestos) |
| 22     | Spring Housing Cover | Mild Steel                  |

#### Dimensions (mm)

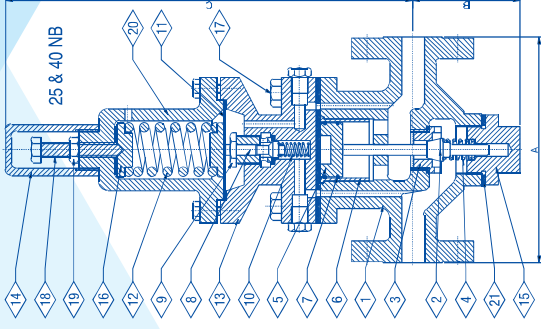
| SIZE | A ±2.0 | B ±5.0 | C ±10.0 |
|------|--------|--------|---------|
| 15   | 160    | 92     | 350     |
| 20   | 160    | 92     | 350     |
| 25   | 200    | 79     | 350     |

#### Features:

- Compact & Versatile design for accurate regulation with widest possible range of flow.
- High flow rate with greater accuracy due to full lift design.
- Stallited trims for high pressure & temperature.
- A positive tight shut off by accurate guiding of pilot & main valve assembly.
- Constant Outlet pressure is accurately maintain without unaffecting by changes in inlet pressure or flow rate.
- Stainless Steel Diaphragm & Harden internal parts provide high performance over long period & trouble free service life.
- Non metallic trims to give dead tight shut off on air & non corrosive gas application

**Note :** \*We Recommend to Provide External Balance Pipe when the Reduced Pressure is Below 55% of the Inlet Pressure

PRV-100 F



#### Dimensions (mm)

| SIZE | A ±2.0 | B ±5.0 | C ±10.0 |
|------|--------|--------|---------|
| 25   | 194    | 79     | 350     |
| 40   | 210    | 102    | 365     |

300#

| SIZE | A ±2.0 | B ±5.0 | C ±10.0 |
|------|--------|--------|---------|
| 25   | 206    | 79     | 350     |
| 40   | 214    | 102    | 365     |

Table H

| SIZE | A ±2.0 | B ±5.0 | C ±10.0 |
|------|--------|--------|---------|
| 25   | 194    | 79     | 350     |
| 40   | 210    | 102    | 365     |

#### Dimensions (mm)

| SIZE | A ±2.0 | B ±5.0 | C ±10.0 |
|------|--------|--------|---------|
| 50   | 254    | 121    | 378     |
| 80   | 311    | 140    | 520     |
| 100  | 347    | 147    | 565     |
| 150  | 433    | 210    | 595     |

150#

| SIZE | A ±2.0 | B ±5.0 | C ±10.0 |
|------|--------|--------|---------|
| 50   | 260    | 121    | 378     |
| 80   | 320    | 140    | 520     |
| 100  | 363    | 147    | 565     |
| 150  | 455    | 210    | 595     |

300#

| SIZE | A ±2.0 | B ±5.0 | C ±10.0 |
|------|--------|--------|---------|
| 50   | 254    | 121    | 378     |
| 80   | 308    | 140    | 520     |
| 100  | 351    | 147    | 565     |
| 150  | 439    | 210    | 595     |

Table H

#### Technical Specifications

|                    |                                                               |
|--------------------|---------------------------------------------------------------|
| Size Range         | 15 mm to 25 mm (Screwed End)<br>25 mm to 150 mm (Flanged End) |
| Pressure Rating    | 150# & 300# / BS 10 Table H                                   |
| Facing             | Raise Face (R.F) , Flat Face (F.F)                            |
| Body M.O.C         | Carbon Steel / Alloy Steel / Stainless Steel.                 |
| Trims              | CF8 / 8M / SS 304 / 316                                       |
| Leakage            | Class IV                                                      |
| Pressure Reduction | 20:1                                                          |
| Min.Differential   | 7 psig.                                                       |

#### Application:

**Steam-Con** PRV-100 Series of Internally Pilot & Piston Operated Pressure Reducing Valve Provide Extremely accurate levels of Pressure Regulation for Steam, air and Industrial gas application. This is accomplished with a highly sensitive pilot which continuously monitors both inlet & outlet Pressure simultaneously. The result is a constant pressure irrespective of erratic inlet pressure of system demand.

#### Ordering / Sizing Information:

- Fluid/State
- Flow Rate
- Inlet Pressure
- Outlet Pressure
- Operating Temp.
- Line Size

Discharge Capacity Chart (PRV-100 T & PRV-100 F) Saturated Steam (kg/hr)

| Inlet Pr.<br>[PSIG] | Outlet<br>Pr.<br>[PSIG] | Nominal Valve Size (mm) |     |     |      |      |      |       |       |     |     |
|---------------------|-------------------------|-------------------------|-----|-----|------|------|------|-------|-------|-----|-----|
|                     |                         | 15                      | 20  | 25  | 40   | 50   | 80   | 70    | 108   | 150 | 250 |
| 10                  | Valve CV                | 5                       | 6.5 | 10  | 18.5 | 32   | 70   | 108   | 250   | 150 | 250 |
|                     | 1                       | 41                      | 83  | 136 | 299  | 529  | -    | -     | -     | -   | -   |
|                     | 3                       | 39                      | 79  | 127 | 282  | 485  | -    | -     | -     | -   | -   |
| 15                  | 5                       | 35                      | 70  | 114 | 251  | 432  | -    | -     | -     | -   | -   |
|                     | 1                       | 52                      | 105 | 171 | 379  | 661  | 1322 | 2358  | -     | -   | -   |
|                     | 3                       | 52                      | 105 | 171 | 379  | 661  | 1322 | 2358  | 4850  | -   | -   |
|                     | 5                       | 48                      | 101 | 163 | 361  | 617  | 1234 | 2248  | 4629  | -   | -   |
|                     | 10                      | 37                      | 75  | 127 | 277  | 485  | -    | -     | -     | -   | -   |
| 20                  | 1                       | 57                      | 114 | 189 | 414  | 727  | 1432 | 2579  | -     | -   | -   |
|                     | 5                       | 61                      | 123 | 202 | 440  | 771  | 1521 | 2755  | 5731  | -   | -   |
|                     | 10                      | 52                      | 110 | 180 | 392  | 683  | 1366 | 2447  | 5070  | -   | -   |
|                     | 15                      | 41                      | 83  | 136 | 299  | 529  | -    | -     | -     | -   | -   |
|                     | 1                       | 61                      | 123 | 202 | 440  | 749  | 1521 | 2645  | -     | -   | -   |
| 25                  | 5                       | 70                      | 136 | 224 | 485  | 859  | 1719 | 3086  | 6393  | -   | -   |
|                     | 10                      | 66                      | 136 | 224 | 485  | 837  | 1697 | 3064  | 6393  | -   | -   |
|                     | 15                      | 61                      | 123 | 198 | 436  | 749  | 1499 | 2711  | 5511  | -   | -   |
|                     | 20                      | 48                      | 92  | 154 | 335  | 573  | -    | -     | -     | -   | -   |
|                     | 1                       | 66                      | 132 | 216 | 485  | 815  | 1631 | 2954  | -     | -   | -   |
| 35                  | 5                       | 74                      | 154 | 251 | 551  | 947  | 1895 | 3439  | 7054  | -   | -   |
|                     | 10                      | 88                      | 176 | 286 | 639  | 1080 | 2160 | 3902  | 7936  | -   | -   |
|                     | 15                      | 88                      | 176 | 286 | 639  | 1080 | 2160 | 3902  | 7936  | -   | -   |
|                     | 25                      | 70                      | 136 | 229 | 507  | 859  | 1719 | 3130  | 6393  | -   | -   |
|                     | 30                      | 52                      | 105 | 171 | 383  | 661  | -    | -     | -     | -   | -   |
| 50                  | 1                       | 74                      | 141 | 242 | 529  | 925  | 1829 | 3306  | -     | -   | -   |
|                     | 5                       | 83                      | 171 | 282 | 617  | 1058 | 2116 | 3813  | 7936  | -   | -   |
|                     | 10                      | 97                      | 193 | 317 | 705  | 1212 | 2402 | 4320  | 8818  | -   | -   |
|                     | 20                      | 114                     | 228 | 379 | 837  | 1432 | 2843 | 5070  | 10581 | -   | -   |
|                     | 30                      | 105                     | 207 | 343 | 749  | 1300 | 2601 | 4629  | 9700  | -   | -   |
| 60                  | 40                      | 79                      | 158 | 264 | 573  | 1014 | 1851 | 3350  | 6834  | -   | -   |
|                     | 45                      | 57                      | 114 | 189 | 414  | 705  | -    | -     | -     | -   | -   |
|                     | 1                       | 74                      | 150 | 246 | 529  | 925  | 1851 | 3328  | -     | -   | -   |
|                     | 5                       | 88                      | 176 | 295 | 639  | 1102 | 2204 | 3990  | 8156  | -   | -   |
|                     | 10                      | 101                     | 246 | 335 | 749  | 1278 | 2535 | 4629  | 9479  | -   | -   |
| 80                  | 20                      | 127                     | 251 | 414 | 903  | 1565 | 3130 | 5731  | 11684 | -   | -   |
|                     | 30                      | 132                     | 260 | 432 | 947  | 1631 | 3262 | 5952  | 11904 | -   | -   |
|                     | 45                      | 105                     | 211 | 348 | 749  | 1300 | 2557 | 4629  | 9479  | -   | -   |
|                     | 50                      | 88                      | 176 | 291 | 639  | 1102 | -    | -     | -     | -   | -   |
|                     | 1                       | 83                      | 163 | 268 | 595  | 1036 | 2050 | 3703  | -     | -   | -   |
| 100                 | 5                       | 92                      | 189 | 308 | 683  | 1168 | 2336 | 4210  | 8818  | -   | -   |
|                     | 10                      | 110                     | 220 | 357 | 793  | 1366 | 2733 | 4850  | 10140 | -   | -   |
|                     | 20                      | 141                     | 277 | 462 | 1014 | 1741 | 3483 | 6172  | 12786 | -   | -   |
|                     | 35                      | 167                     | 334 | 551 | 1212 | 2094 | 4188 | 7495  | 15431 | -   | -   |
|                     | 50                      | 154                     | 308 | 507 | 1124 | 1917 | 3857 | 6834  | 14329 | -   | -   |
| 120                 | 65                      | 119                     | 233 | 388 | 837  | 1455 | 2711 | 4850  | 10140 | -   | -   |
|                     | 70                      | 97                      | 198 | 326 | 705  | 1234 | -    | -     | -     | -   | -   |
|                     | 1                       | 88                      | 176 | 291 | 639  | 1102 | 2204 | 3924  | -     | -   | -   |
|                     | 5                       | 101                     | 198 | 330 | 727  | 1234 | 2491 | 4409  | 9259  | -   | -   |
|                     | 10                      | 114                     | 229 | 388 | 859  | 1477 | 2932 | 5290  | 10802 | -   | -   |
| 150                 | 25                      | 149                     | 295 | 485 | 1080 | 1851 | 3703 | 6613  | 13668 | -   | -   |
|                     | 35                      | 185                     | 374 | 617 | 1366 | 2336 | 4629 | 8377  | 17195 | -   | -   |
|                     | 50                      | 202                     | 405 | 661 | 1477 | 2535 | 5070 | 9038  | 18738 | -   | -   |
|                     | 70                      | 176                     | 352 | 573 | 1278 | 2204 | 4409 | 7936  | 16313 | -   | -   |
|                     | 85                      | 136                     | 264 | 440 | 970  | 1675 | 3042 | 5511  | 11684 | -   | -   |
| 200                 | 90                      | 110                     | 224 | 370 | 815  | 1388 | -    | -     | -     | -   | -   |
|                     | 5                       | 110                     | 220 | 361 | 793  | 1366 | 2733 | 4850  | -     | -   | -   |
|                     | 10                      | 127                     | 251 | 418 | 925  | 1587 | 3152 | 5731  | 11684 | -   | -   |
|                     | 20                      | 154                     | 308 | 507 | 1124 | 1940 | 3880 | 7054  | 14329 | -   | -   |
|                     | 30                      | 185                     | 409 | 595 | 1322 | 2292 | 4629 | 8156  | 16975 | -   | -   |
| 250                 | 50                      | 229                     | 462 | 771 | 1675 | 2887 | 5731 | 10361 | 21384 | -   | -   |
|                     | 75                      | 220                     | 440 | 727 | 1609 | 2755 | 5511 | 9920  | 20502 | -   | -   |
|                     | 105                     | 149                     | 295 | 485 | 1080 | 1851 | 3086 | 5511  | 11463 | -   | -   |
|                     | 110                     | 123                     | 242 | 396 | 881  | 1521 | -    | -     | -     | -   | -   |
|                     | 110                     | 123                     | 242 | 396 | 881  | 1521 | -    | -     | -     | -   | -   |

Quick Sizing for Pressure Reducing Valve

Discharge Capacity Chart (PRV-100 T & PRV-100 F) Saturated Steam (kg/hr)

| Inlet Pr.<br>[PSIG] | Outlet<br>Pr.<br>[PSIG] | Nominal Valve Size (mm) |      |      |      |      |       |       |       |  |  |
|---------------------|-------------------------|-------------------------|------|------|------|------|-------|-------|-------|--|--|
|                     |                         | 15                      | 20   | 25   | 40   | 50   | 80    | 100   | 150   |  |  |
| 150                 | Valve CV                | 5                       | 6.5  | 10   | 18.5 | 32   | 70    | 108   | 250   |  |  |
|                     | 1                       | 88                      | 176  | 291  | 639  | 1102 | 2204  | 3924  | -     |  |  |
|                     | 5                       | 110                     | 220  | 365  | 793  | 1388 | 2755  | 5070  | -     |  |  |
|                     | 10                      | 132                     | 268  | 440  | 970  | 1675 | 3306  | 5952  | 12345 |  |  |
|                     | 15                      | 149                     | 340  | 485  | 1080 | 1851 | 3703  | 6613  | 13668 |  |  |
|                     | 25                      | 180                     | 365  | 595  | 1300 | 2248 | 4409  | 8156  | 16754 |  |  |
|                     | 50                      | 255                     | 510  | 815  | 1807 | 3130 | 6172  | 11243 | 23147 |  |  |
|                     | 75                      | 291                     | 575  | 970  | 2116 | 3637 | 7275  | 13006 | 26895 |  |  |
|                     | 100                     | 264                     | 530  | 881  | 1917 | 3306 | 6613  | 11904 | 24250 |  |  |
|                     | 120                     | 220                     | 440  | 727  | 1609 | 2755 | 5290  | 9479  | 19400 |  |  |
| 180                 | 135                     | 167                     | 340  | 551  | 1234 | 2116 | 3483  | 6172  | 12786 |  |  |
|                     | 140                     | 136                     | 285  | 462  | 1014 | 1741 | -     | -     | -     |  |  |
|                     | 1                       | 141                     | 285  | 462  | 1014 | 1763 | 3350  | 6393  | -     |  |  |
|                     | 25                      | 189                     | 380  | 617  | 1366 | 2358 | 4629  | 8597  | 17415 |  |  |
|                     | 50                      | 268                     | 530  | 881  | 1940 | 3350 | 6613  | 12125 | 24911 |  |  |
|                     | 75                      | 326                     | 660  | 1080 | 2358 | 4078 | 8156  | 14550 | 30202 |  |  |
|                     | 100                     | 343                     | 685  | 1146 | 2491 | 4298 | 8597  | 15431 | 31745 |  |  |
|                     | 120                     | 313                     | 620  | 1036 | 2270 | 3902 | 7936  | 14109 | 28879 |  |  |
|                     | 150                     | 242                     | 485  | 793  | 1741 | 2998 | 5511  | 10581 | 21604 |  |  |
|                     | 165                     | 180                     | 360  | 595  | 1300 | 2226 | 3615  | 6834  | 14109 |  |  |
| 200                 | 1                       | 141                     | 321  | 485  | 1036 | 1807 | 3615  | 6393  | -     |  |  |
|                     | 25                      | 202                     | 396  | 661  | 1432 | 2469 | 4850  | 8818  | 18297 |  |  |
|                     | 50                      | 273                     | 595  | 903  | 1984 | 3417 | 6834  | 12345 | 25131 |  |  |
|                     | 75                      | 339                     | 685  | 1124 | 2469 | 4012 | 8597  | 15431 | 31525 |  |  |
|                     | 100                     | 379                     | 750  | 1256 | 2755 | 4850 | 9479  | 16975 | 35098 |  |  |
|                     | 120                     | 365                     | 730  | 1212 | 2667 | 4629 | 9259  | 16534 | 33950 |  |  |
|                     | 150                     | 317                     | 640  | 1036 | 2270 | 3924 | 7936  | 14109 | 29100 |  |  |
|                     | 185                     | 189                     | 380  | 617  | 1366 | 2380 | 3791  | 7275  | 14770 |  |  |
|                     | 15                      | 167                     | 334  | 551  | 1212 | 2116 | 4188  | 7583  | -     |  |  |
|                     | 20                      | 299                     | 595  | 992  | 2182 | 3747 | 6850  | 13447 | 27777 |  |  |
| 250                 | 75                      | 365                     | 725  | 1212 | 2645 | 4629 | 9126  | 16313 | 33729 |  |  |
|                     | 100                     | 427                     | 858  | 1410 | 3086 | 5290 | 10581 | 19179 | 39240 |  |  |
|                     | 150                     | 462                     | 902  | 1499 | 3284 | 5731 | 11243 | 20502 | 42988 |  |  |
|                     | 175                     | 414                     | 836  | 1366 | 3020 | 5290 | 10361 | 18738 | 38579 |  |  |
|                     | 200                     | 361                     | 725  | 1190 | 2601 | 4409 | 8818  | 16093 | 32406 |  |  |
|                     | 220                     | 291                     | 572  | 947  | 2094 | 3637 | 6393  | 11463 | 26013 |  |  |
|                     | 235                     | 207                     | 409  | 683  | 1499 | 2579 | 4100  | 7230  | 14770 |  |  |
|                     | 2                       | 189                     | 378  | 617  | 1366 | 2358 | -     | -     | -     |  |  |
|                     | 50                      | 313                     | 616  | 1036 | 2248 | 3880 | 7715  | 13888 | -     |  |  |
|                     | 75                      | 388                     | 770  | 1278 | 2821 | 4850 | 9700  | 17415 | 35934 |  |  |
| 300                 | 100                     | 462                     | 925  | 1565 | 3350 | 5731 | 11463 | 20722 | 42547 |  |  |
|                     | 150                     | 551                     | 1100 | 1807 | 3990 | 6834 | 13668 | 24690 | 50704 |  |  |
|                     | 175                     | 551                     | 1100 | 1807 | 3990 | 6834 | 13668 | 24690 | 50704 |  |  |
|                     | 200                     | 507                     | 1035 | 1697 | 3747 | 6393 | 12786 | 23147 | 47618 |  |  |
|                     | 220                     | 462                     | 946  | 1565 | 3417 | 5952 | 11904 | 21163 | 43650 |  |  |
|                     | 250                     | 401                     | 792  | 1322 | 2910 | 5070 | 9479  | 16975 | 34831 |  |  |
|                     | 285                     | 238                     | 484  | 793  | 1741 | 2976 | 4188  | 7715  | 15872 |  |  |
|                     | 40                      | 282                     | 572  | 925  | 2028 | 3549 | 11022 | 19840 | -     |  |  |
|                     | 90                      | 440                     | 880  | 1477 | 3218 | 6393 | 13447 | 24911 | 49381 |  |  |
|                     | 120                     | 529                     | 1055 | 1763 | 3880 | 6613 | 13447 | 24911 | 49381 |  |  |
| 350                 | 150                     | 595                     | 1188 | 1984 | 4342 | 7495 | 14990 | 26895 | 55554 |  |  |
|                     | 180                     | 639                     | 1275 | 2116 | 4629 | 8156 | 16093 | 28879 | 59522 |  |  |
|                     | 210                     | 639                     | 1254 | 2094 | 4629 | 7936 | 15872 | 28218 | 58200 |  |  |
|                     | 240                     | 595                     | 1188 | 1962 | 4298 | 7495 | 14770 | 26675 | 54672 |  |  |
|                     | 270                     | 529                     | 1055 | 1741 | 3835 | 6613 | 13227 | 23588 | 48500 |  |  |
|                     | 300                     | 436                     | 880  | 1455 | 3174 | 5511 | 10140 | 18297 | 37477 |  |  |